Canonical graded pieces of the literal JV quotient

Result. Lean proves that the first printed Joux–Vitse homogeneous F101 source ideal has canonical Hilbert function1,3,3,1,0 thereafter. A canonical degree piece is zero exactly when its degree is at least4. This closes the prior finite per-degree identification gap, supplementing the already-proved actual quotient basis and total dimension8. No native Hilbert or Gröbner result is assumed. [pcc:unverified]

Canonical pieces	P_d / ker(actual quotient map restricted to P_d) Kernel = I intersect P_d; firstisomorphism to actual image
Derived graded basis	Degree: 0 1 2 3 Representatives:1 x,y,h xh,yh,h² h³
Exact dimensions	Hilbert function:1,3,3,1,0,0,...; first zero degree4 Total scalar dimension/length8; HF(0)=1 differs from8

Actual canonical construction. Let P=F101[x,y,h] and I be the unchanged ideal of the three literal homogenized coordinate equations. Lean proves I homogeneous for Mathlib standard grading. For each natural d, restrict the actual quotient map P→P/I to Mathlib homogeneousSubmodule of degree d. Prove its kernel is exactly I intersected with that submodule. The canonical quotient piece P_d/(I∩P_d) is linearly equivalent to the actual image by the first isomorphism theorem. These definitions precede comparison with a proposed basis. [pcc:unverified]

Derived comparison. The unchanged actual quotient proof supplies a basis and24 checked coordinate multiplication relations. Kernel-checked finite tables show every nonzero coefficient raises degree by one. Coordinate/power/monomial induction then proves that every actual homogeneous polynomial image lies in the matching-degree basis span. Conversely, every selected basis representative is an actual homogeneous polynomial image. The actual image equals that span; inherited linear independence gives its dimension, with no Hilbert/basis/regularity conclusion supplied as a premise. [pcc:unverified]

Control and boundaries. HF(0)=1 while total scalar dimension and scalar module length are8; Lean directly negates their equality. The exact Hilbert cutoff4 is separate from maximal internal F4 processing degrees. Generic JV Assumption2, FGHR generic/n≥3 scope, larger-system cohorts, whole-curve order/scalar relations and native-C refinement remain open. The theorem does not claim a generic graded-module Hilbert polynomial API or a new source counterexample. [pcc:unverified]

Checks and provenance. Whole new project gate26/0 failures/0 trusted native; only propext, Classical.choice and Quot.sound. Complete712-character written binding288fa232f61b729eee2a64e17f2741efcf7c144c019684d0bc676e5fcd2db8b6, independently parsed1complete/0prefix/0unresolved. Native parsing/hash evidence is procedural, not a kernel refinement theorem. Five build snapshots, including four failures, remain: 62files/32objects/11452-byte lossless XZ plus five final gate files. Frozen old sources unchanged. [pcc:unverified]

Sources. Joux–Vitse author21-page manuscript, full selectedp7literal source equations andp9Assumption2/generic-degree discussion read and visually inspected; PDFSHA6f3268a0…. No fullpaper-read or novelty claim. Prior actual quotient/dual/cutoff/source-field work credited. Official Mathlib homogeneous modules/ideals/firstisomorphism/basis dimensions read; its standard grading instance is explicitly local as documented. HilbertPoly explicitly TODOs generic graded-module theory, which is not assumed here. KB/Conductor unavailable; primary/prior/official fallback. Lean4.35.0/Mathlib56a00585…, standard defaults; no holes/custom axioms/native proof evaluation/flags/limits/waivers/new crypto experiment/spend/upload. Signed standing remains unverified. Source-linked proof map/pins accompany the report. [pcc:unverified]